CHEMISTRY | ADAPTLY

QUANTITATIVE CHEMISTRY – PERCENTAGE YIELD

KEY VOCABULARY

Yield: the mass of product actually made.

Theoretical yield: the maximum mass that could be made.

Percentage yield: actual ÷ theoretical × 100.

Reversible reaction: a reaction that lowers yield as it goes both ways.

COMMON MISCONCEPTIONS

~~100% yield is always possible.~~

✓ Product is lost, so yield is usually below 100%.

~~Yield and atom economy are the same.~~

✓ Yield = how much; atom economy = how efficient.

~~Low yield means the equation is wrong.~~

✓ Losses happen during transfer and side reactions.

PERCENTAGE YIELD

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

CONVERSION REMINDER

Yield is never more than 100%

1 CALCULATE YIELD

Theoretical yield 50 g; actual yield 40 g. **Calculate** the percentage yield. **[2 marks]**

Answer: ______________________ %

2 REARRANGE

% yield is 80% and theoretical yield is 25 g. **Calculate** the actual yield. **[2 marks]**

Answer: ______________________ g

3 WHY NOT 100%

Give two reasons why the yield is often less than 100%. **[2 marks]**

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4 REVERSIBLE

Explain why a reversible reaction reduces the yield. **[2 marks]**

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5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Percentage yield = (_____ ÷ _____ yield) × 100.

actual | theoretical | 100 | 10

____

6 EVALUATE

Suggest why a high percentage yield is important to industry. **[1 mark]**

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TIP: *% yield compares what you got with the maximum you could have got.*

✓ Think: Why can losing product on the filter paper lower your yield?